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Multimodal Imaging and Dark-Adapted Chromatic Perimetry in BEST1 Vitelliform Macular Dystrophy: Identification of Outcome Measurements

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Objective: To characterize longitudinal multimodal imaging characteristics of Best vitelliform macular dystrophy (BVMD) and describe novel outcome measurements for future clinical trials.

Design: A single-center retrospective cohort study.

Subjects: A total of 36 eyes of 18 patients with genetically and clinically confirmed diagnoses of BVMD were evaluated.

Methods: Patients underwent complete ophthalmic examination and multimodal imaging including spectral-domain OCT and short-wavelength autofluorescence. Dark-adapted chromatic (DAC) perimetry was obtained in 8 of 18 patients.

Main Outcome Measures: Longitudinal changes in lesion width, area, and measures of retinal thickness, including outer nuclear layer (ONL) thickness, were evaluated as primary outcome measures. The changes were measured at every visit to the ophthalmology clinic, with a difference between the first and most recent individual's visit of 76.56 ± 33.36 months (range 10–123 months), on average. Scotopic sensitivity as measured by DAC perimetry was compared within and outside the lesion.

Results: Lesion width increased with progressive disease stage from the vitelliform stage (1.345 ± 0.218 mm) through the vitelliruptive stage (2.913 ± 0.893 mm) before decreasing in the atrophic stage (2.287 ± 0.456 mm). Central macular thickness increased between the vitelliform (0.277 ± 0.132 mm) and the pseudohypopyon stage (0.334 ± 0.055 mm) before decreasing through the vitelliruptive (0.288 ± 0.085 mm) and atrophic stages (0.236 ± 0.020 mm). Measurements of ONL thickness at the temporal and nasal borders of the lesion demonstrated a significant difference over time ($P = 0.001$). Dark-adapted chromatic perimetry showed significant differences in scotopic sensitivity within the lesion compared with outside the lesion and compared with values for normal control subjects ($P < 0.05$).

Conclusions: Features of multimodal imaging, including measurements of ONL thickness along the temporal and nasal borders of the lesion, as well as scotopic sensitivity as measured by DAC perimetry, may serve as valuable outcome measurements for treatment trials for BVMD.

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Vitelliform macular lesions are a common feature of many retinal disorders.[1–3] Pathogenic variants in *BEST1* (Online Mendelian Inheritance in Man 607854), the gene encoding a basolateral transmembrane protein expressed in the retinal pigment epithelium (RPE) cells, are responsible for Best vitelliform macular dystrophy (BVMD).[4,5] Although its function remains debated, there is evidence that BEST1 regulates Ca^{2+} homeostasis and also acts as an anion channel permeable to chloride and HCO_3^-.[5–10] Studies performed in naturally occurring *BEST1* knockout canine animal models suggest that protein-altering variants in this gene modify both the microvilli of RPE surface and the ionic environment of the subretinal space.[8] This alteration is believed to be the reason for subretinal fluid accumulation between the outer segments of photoreceptor cells and the apical surface of the RPE cells and is responsible for the initial manifestation of BVMD—the vitelliform lesion.[4,8]

Initially, patients presenting with vitelliform lesions have relatively preserved visual acuity and mild symptoms.[4,11] However, the persistence of fluid may be detrimental to photoreceptor–RPE interactions, and disease progression may occur.[4,8] The vitelliform stage is then followed by the

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pseudohypopion, vitelliruptive, and atrophic stages.[4,11] The symptoms are more evident in the vitelliruptive stage, and a more dramatic visual acuity change is seen in the latter stage.[11] Nevertheless, not every patient will progress through every stage of the disease; moreover, eye asymmetry is a common feature of BVMD.[4,11–13]

BEST1 stands as an excellent candidate for gene augmentation therapy, and many efforts have been made to develop a treatment for *BEST1*-related macular dystrophies.[14–17] In fact, the autosomal recessive canine disease model has shown promising results.[14] Several studies have shown the potential for a similar approach to treat human disease caused by *BEST1* variants.[15–17] In cases where a dominant negative or gain-of-function effect is present, a modified approach can potentially be used.[17,18] As for clinical trials, however, the phenotypic variability associated with BVMD makes it challenging to determine appropriate outcome measures. To date, few studies have characterized longitudinal biomarkers of disease.[19,20] More recently, Cideciyan et al have proposed the use of a modified Humphrey visual field analyzer to monitor functional changes in photoreceptors.[21] Herein, we report the features of disease progression in BVMD using multimodal imaging, and we describe findings using a novel functional biomarker of disease that might be useful in future clinical trials.

Methods

We conducted a retrospective review of 18 patients who were evaluated at the Columbia University Irving Medical Center with a confirmed clinical and molecular genetic diagnosis of BVMD. Each patient underwent a comprehensive ophthalmic examination, including measurement of best-corrected visual acuity and dilated fundus examination. After dilation, patients underwent multimodal imaging, including color fundus photography, spectral-domain OCT (SD-OCT), and short-wavelength autofluorescence (SW-AF). Both SD-OCT and SW-AF images were obtained using a confocal scanning laser ophthalmoscope (Spectralis HRA + OCT, Heidelberg Engineering), with SW-AF images obtained at an excitation of 488 nm with barrier filter transmitted light from 500 to 680 nm. The study protocol was approved by the Columbia University's Institutional Review Board (AAAR8743), and all procedures adhered to the tenets of the Declaration of Helsinki. Informed consent was deferred by our institutional review board due to the retrospective nature of the study and minimal risk conferred to patients.

Multimodal Imaging Analysis

To monitor disease progression, we compared the results from each visit at which imaging was obtained. In the SW-AF images, measurements of the width at the fovea and area of the vitelliform lesion were taken using the built-in caliper tool on the Heidelberg Explorer. To ensure consistency, these measurements were obtained after aligning images based on the distribution of blood vessels. Using SD-OCT scans, measurements of lesion width at the fovea, central macular thickness (CMT), and central lesion thickness were recorded when available. The thickness of the outer nuclear layer (ONL) was measured at 5 points, including the center of the lesion, the margins of the lesion, and 500 μms from the border of the lesion. Measurements of retinal thickness and lesion width were performed using the built-in caliper tool on the Heidelberg Explorer. If available, measurements were taken using follow-up scans that used real-time tracking to align with prior imaging. When tracked scans were not available, measurements were taken at the center of the foveal pit from both visits.

Dark-Adapted Chromatic Perimetry

After administration of topical 1% tropicamide and 2.5% phenylephrine eye drops, the patient was placed in a dark room for 40 minutes with a black patch over the eye to be tested. A dark-adapted chromatic (DAC) perimeter (Medmont International Pty Ltd) was used to measure scotopic sensitivity. The test eye was aligned in the infrared viewing window, and the position of the pupil was monitored throughout the examination. Corrective lenses were placed in front of the patient's eyes when necessary to correct for refractive errors and the viewing distance. A 505-nm wavelength (cyan) stimulus (dynamic range 0–75 decibel [dB]) and a 625-nm wavelength (red) stimulus (dynamic range 0–50 dB) were used to test 52 locations separated by 2 each in the central 48° of the visual field of each patient (Fig S3, available at www.ophthalmologyscience.org). A stimulus size of 1.73° (equivalent to Goldmann size V) was used with a stimulus duration of 200 ms. The DAC perimeter uses a black bowl and a background luminance of <0.0001 cd/m^2. Seventeen control subjects with an average age of 32.5 years (range 22–47 years) and no past ocular history and visual acuity of 20/20 in both eyes were used for comparison.

Measurements of dark-adapted sensitivity were extracted from the DAC perimeter, and the numeric maps in dB were manually aligned on top of 55° SW-AF images with the fixation point placed over the fovea and the blind spot over the optic disc. The location of the fovea was determined using SD-OCT images after the anatomy of the retinal vessels was used to align the near-infrared reflectance component of the SD-OCT and SW-AF images. Care was taken to transpose retinal view from the visual field view by inverting and flipping the visual field points to correspond to the retinal view.

Statistical Analysis

Descriptive statistics were used to describe baseline lesion characteristics on multimodal imaging. A Pearson correlation coefficient was calculated to determine the association between lesion measurements obtained on SD-OCT and SW-AF. Paired sample *t* tests and 1-way analysis of variance were used to determine longitudinal change over time. All the measurements were retaken, $\geq$2 weeks apart, by the same observer (J.K.O.), and intraclass correlation coefficients were calculated to determine the reliability of intergrading variability. For measurements of retinal function, Welch *t* test was used to compare retinal function within the lesion and outside the lesion in patients with disease as well as between patients with and without disease. Threshold for statistical significance was set at a *P* value of 0.05. All statistical analysis was performed using R statistical software version 3.61 (R Foundation for Statistical Computing).

Results

Demographics

A total of 18 individuals (30 eyes) were included in the study; 15 of the 18 underwent longitudinal multimodal imaging with an average follow-up of 76.56 ± 33.36 months (range 10–123 months) between the initial and most recent evaluation (Table 1). Eyes were assigned to clinical stages by 2 authors (J.R.L.C. and J.K.O.) based on previously described features in SD-OCT scans. At the

Table 1. Baseline Demographic Features of Eyes with Best Vitelliform Macular Dystrophy

Patient ID	Age at Initial Visit (Yrs)	Follow-Up Interval (Mos)	Disease Stage at Initial Visit (OD/OS)	Disease Stage at Recent Visit (OD/OS)	BCVA (OD/OS)	Variant
P1	43	42	II/I	II/I	20/25 & 20/20	c.884T > C:(p.Ile295Thr)
P2	32	94	IV/IV	IV/IV	20/20 & 20/20	c.727G > A:(p.Ala243Thr)
P3	5	91	III/III	IV/III	20/80 & 20/25	c.28G > A:(p.Ala10Thr)
P4	45	59	IV/V	IV/V	20/50 & 20/60	c.89A > G:(p.Lys30Arg)
P5	16	82	IV/IV	IV/IV	20/20 & 20/20	c.887A > G:(p. Asn296Ser)
P6	55	82	III/III	IV/IV	20/100 & 20/80	c.887A > G:(p. Asn296Ser)
P7	30	97	III/II	III/II	20/20 & 20/150	c.727G > A:(p.Ala243Thr)
P8	44	10	IV/IV	IV/IV	20/50 & 20/50	c.877C > A:(p.Gln293Lys)
P9	6	120	IV/II	IV/IV	20/20 & 20/20	c.701T > C:(p.Leu234Pro)
P10	13	26	II/II	II/II	20/20 & 20/20	c.653G > A:(p.Arg218His)
P11	52	104	III/III	III/III	20/150 & 20/60	c.653G > A:(p.Arg218His)
P12	47	123	IV/IV	IV/IV	20/80 & 20/70	c.253T > C:(p.Tyr85His)
P13	55	80	IV/IV	V/V	20/70 & 20/70	c.89A > G:(p.Lys30Arg)
P14	72	94	II/IV	V/V	20/20 & 20/150	c.896G > A:(p.Gly299Glu)
P15	53	45	V/IV	V/V	20/100 & 20/50	c.253T > C:(p.Tyr85His)
P16	57	X	V/IV	NA	20/400 & 20/80	c.652C > T:(p.Arg218Cys)
P17	45	X	V/V	NA	20/400 & 20/400	c.218T > A:(p.Asn73Thr)
P18	43	X	IV/IV	NA	20/30 & 20/20	c.652C > T:(p.Arg218Cys)

BCVA = best-corrected visual acuity; NA = not available; OD = right eye; OS = left eye.
Disease stage.
I – Previtelliform.
II – Vitelliform.
III – Pseudohypopyon.
IV – Vitelliruptive.
V – Atrophic.

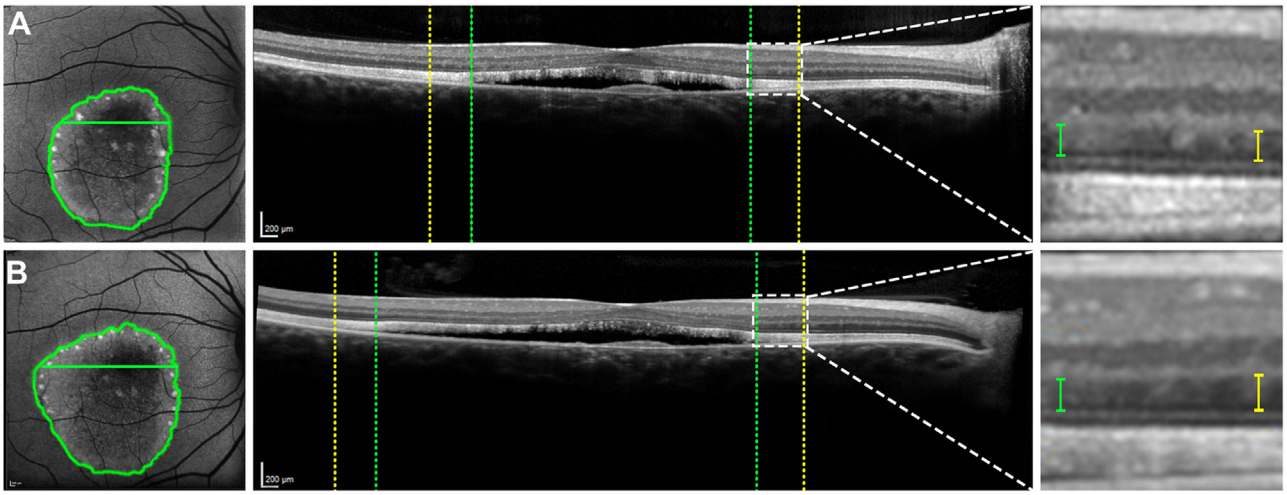


Figure 1. Progression of Best vitelliform macular dystrophy on multimodal imaging. Patient 2 (P2). **A,** Initial presentation. **B,** Follow-up examination 94 months later. Foveal lesion width and lesion area measured in SW-AF images (**A, B** left; solid green lines). Lesion width was also measured on SD-OCT scans (**A, B** center panel). The nasal and temporal borders of the lesion are demarcated on SD-OCT (dotted green lines), as are the points 500 µms from the nasal and temporal borders (dotted yellow lines). Outer nuclear layer thickness measurements at the nasal border of the lesion (green bracket) and 500 µm from the nasal border of the lesion (yellow bracket) are demonstrated in the inlays (white box). Comparison of **A** and **B** shows an increase in lesion width and area in the SW-AF images, an increase in lesion width in the SD-OCT image, and a slight increase in thickness of the outer nuclear layer. SW-AF = short-wavelength autofluorescence; SD-OCT = spectral-domain OCT.

initial evaluation, 1 eye was found to be in the previtelliform state, 6 eyes were noted to be in the vitelliform stage, 7 eyes were first evaluated at the pseudohypopyon stage, 14 eyes were found to be in the vitelliruptive stage, and 2 eyes were in the atrophic stage. Symmetry in lesion staging at the initial evaluation was seen in 9 of 15 individuals (18 of 30 eyes) and in 11 individuals at the most recent examination (22 of 30 eyes). A total of 9 of 30 eyes showed disease progression as defined by the clinical stage over the course of follow-up.

Correlations of Multimodal Imaging

Longitudinal analysis of lesion width on SD-OCT was performed for 29 eyes of 15 individuals. At the initial examination, the average width of the lesion was 2452.8 ± 0.885 mm (range 1.086–4.728 mm). There was a median increase in lesion width of 0.105 mm per year (Fig 1). Comparison of measurements on SD-OCT and SW-AF demonstrated a strong linear correlation between measurements ($r = 0.95$) (Fig S1, available at www.ophthalmologyscience.org). Analysis of lesion area over time was also performed, demonstrating an average annual increase in the size of the lesion by 0.93 ± 2.16 mm^2. Comparison of lesion area on SW-AF with lesion width on SD-OCT ($r = 0.91$) and SW-AF ($r = 0.97$), respectively, demonstrated a significant linear correlation ($P < 0.01$).

Characteristics of Lesion Size

Analysis of lesion width in the foveal region in both SD-OCT and SW-AF images demonstrated that average lesion sizes differed depending on the clinical stage ($P < 0.001$, 1-way analysis of variance). Overall, lesion width increased with disease stage (stages II through IV), with the exception of lesions seen in atrophic disease (stage V), which were smaller than those seen in the pseudohypopyon stage (Table 2). In contrast, the average change in lesion size decreased with increasing disease stage, with the greatest increases in lesion width seen in patients who were first seen in the vitelliform stage; however, these differences were not statistically significant on either SD-OCT or SW-AF.

Subanalysis was performed to determine whether there was a difference between patients who had progression of disease stage (n = 9 eyes) compared with those that remained at the same clinical stage (n = 21). Overall, patients who had progression of disease stage had a greater average increase in lesion size (0.216 ± 0.383 mm on SD-OCT and 0.238 ± 0.538 mm on SW-AF) compared with those that remained at the same disease stage (0.056 ± 0.111 mm on SD-OCT and 0.062 ± 0.098 mm on SW-AF); however, these differences were not statistically significant ($P = 0.09$ and 0.16, t test). Longitudinal changes in lesion diameter on SD-OCT and lesion area on SW-AF at multiple time points are illustrated for 17 eyes of 9 individuals who had >2 visits. Patients P3, P9, P13, and P14 were found to have progression of disease stage in either 1 or both eyes (Fig 2).

Characteristics of Lesion and Retinal Thickness

Analysis of thickness measurements demonstrated several consistent trends. Average CMT increased as the disease progressed from the vitelliform (0.277 ± 0.132 mm) to pseudohypopyon stage (0.334 ± 0.055 mm) before decreasing with further progression to the vitelliruptive (0.288 ± 0.085 mm) and atrophic stages (0.236 ± 0.020 mm). Similar changes were noted with

Table 2. Retinal Features of Best Vitelliform Macular Dystrophy as Measured by OCT

Disease Stage	Lesion Width (μm)	Central Macular Thickness (μm)	Central Lesion Height (μm)	Central ONL Thickness (μm)	Temporal ONL Thickness (μm)	Nasal ONL Thickness (μm)	Temporal ONL Thickness + 500 (μm)	Nasal ONL Thickness +500 (μm)
Stage II – Vitelliform	1345.8 ± 218.4	277.3 ± 132.3	212.4 ± 106.3	61.3 ± 29.2	80.9 ± 38.0	85.9 ± 34.2	73.6 ± 41.3	78.0 ± 36.3
Stage III – Pseudohypopyon	2527.1 ± 347.5	334.1 ± 54.9	245.6 ± 78.1	63.9 ± 20.5	64.9 ± 13.8	75.4 ± 12.6	63.3 ± 18.5	76.7 ± 14.9
Stage IV – Vitelliruptive	2913.7 ± 893.8	287.6 ± 84.9	206.5 ± 78.4	62.3 ± 24.5	52.0 ± 11.0	54.1 ± 18.8	50.0 ± 13.5	58.5 ± 15.0
Stage V – Atrophic	2287.5 ± 456.1	236.0 ± 19.8	151.0 ± 43.8	36.0 ± 7.1	66.0 ± 24.8	60.5 ± 17.7	71.5 ± 57.3	68.5 ± 29.0

Disease Stage	ΔLesion Width/Yr (μm)	ΔCentral Macular Thickness/Yr	ΔCentral Lesion Height/Yr	ΔCentral ONL Thickness/Yr	ΔTemporal ONL Thickness/Yr	ΔNasal ONL Thickness/Yr	ΔTemporal ONL Thickness ± 500 μm/Yr	ΔNasal ONL Thickness ±500 μm/Yr
Stage II – Vitelliform	188.2 ± 473.4	−13.6 ± 27.6	−10.2 ± 21.5	−2.1 ± 4.2	2.2 ± 6.3	−0.9 ± 6.0	2.5 ± 6.4	0.7 ± 3.8
Stage III – Pseudohypopyon	139.9 ± 210.9	−0.4 ± 7.6	0.1 ± 8.0	−0.9 ± 2.0	−0.3 ± 3.6	−1.6 ± 2.5	−0.3 ± 3.6	−0.9 ± 2.4
Stage IV –Vitelliruptive	67.1 ± 82.7	−3.2 ± 25.6	−0.9 ± 16.1	−1.9 ± 5.4	−0.7 ± 5.6	2.0 ± 3.1	0.5 ± 8.5	1.4 ± 4.7
Stage V – Atrophic	12.6 ± 4.7	2.8 ± 11.1	3.7 ± 7.5	−2.4 ± 4.9	−1.9 ± 0.4	−1.2 ± 3.3	−2.7 ± 8.1	−0.6 ± 3.4

ONL = outer nuclear layer.

measurements of both the lesion height and the ONL centrally, nasally, and temporally along the borders of the lesion (Table 2). Despite these trends, 1-way analysis of variance did not demonstrate a significant difference between CMT ($P = 0.19$), central lesion height ($P = 0.39$), or central ONL thickness ($P = 0.25$) among disease stages. However, measurements of the ONL along the temporal ($P < 0.001$) and nasal borders ($P < 0.001$) of the lesion as well as 500 μm from the temporal ($P = 0.001$) and nasal borders ($P = 0.002$) were found to be significantly different among stages. Post hoc analysis demonstrated significant differences between all stages except stage IV compared with stage V.

Change in thickness over time was also determined, with the greatest changes seen in patients with vitelliform disease, who demonstrated average annual changes in CMT of 0.014 ± 0.028 mm and central lesion height of 0.010 ± 0.022 mm. In contrast, progression of CMT, central lesion height, and ONL thickness centrally and at previously described anatomical landmarks was minimal, with <5 μm of change per year in any of those parameters across all stages (Table 2) (Fig S2, available at www.ophthalmologyscience.org).

Scotopic Sensitivity

The DAC perimetry was performed on 8 individuals (P11–P18; Table 1). All 8 patients were tested using a grid with 52 locations, as shown in Figure 3. The right eye of each patient was used for analysis. Rod-mediated sensitivity was determined by comparing the sensitivity to the 505 nm (cyan) stimulus at each location to the sensitivity to the 625 nm (red) stimulus as previously described in the literature.[22,23] All analyzed points had a difference in cyan-red sensitivity >10 dB, which is above the 5 dB threshold attributed to cone sensitivity. The numeric maps (dB values) were overlaid on the 55° SW-AF images for the patients with the fixation point placed over the fovea (Fig 3). Scotopic sensitivity at locations within and outside the lesion demonstrated significantly different sensitivity values in 7 of 8 (P12–P18) patients ($P < 0.05$). Additional analysis was performed to compare scotopic sensitivity within the lesion of patients and at the corresponding locations in normal control subjects using Welch t test. In 6 of the 8 patients (P12–P17), a statistically significant difference in scotopic sensitivity was noted between patients and the controls ($P < 0.05$).

Discussion

The treatment of any progressive and degenerative macular disorder requires a rigorous understanding of natural history and, subsequently, the delineation of clear outcome measurements for clinical trials. Best vitelliform macular dystrophy is a disease that is known to have distinct morphological and structural features that have been previously described on multimodal imaging.[4,24,25] Despite the existence of well-accepted classifications of the stages of disease in BVMD, there is limited information on longitudinal changes in disease over time and limited biomarkers to

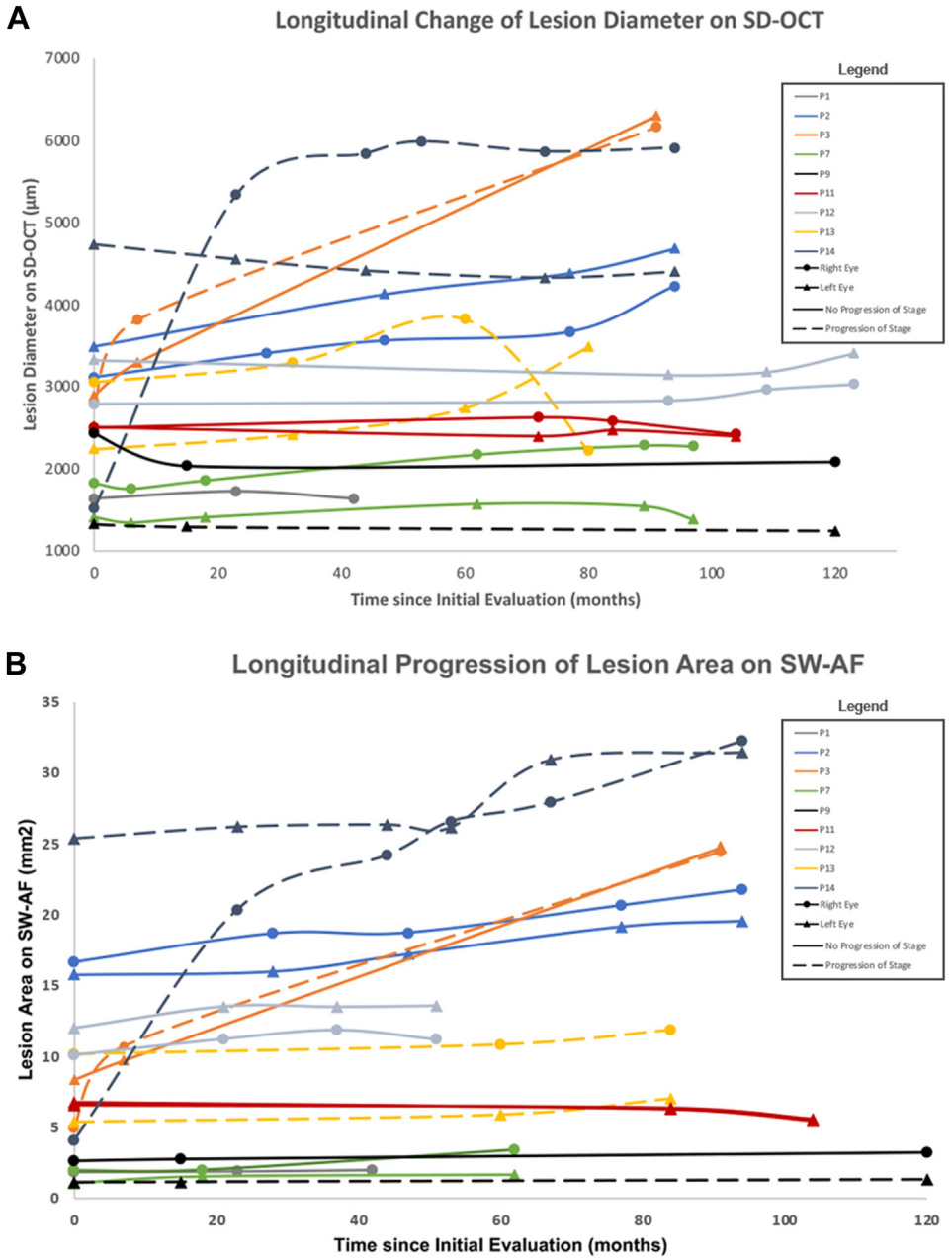


Figure 2. Longitudinal change of vitelliform lesion diameter and area on multimodal imaging. **A,** Change in vitelliform lesion diameter on SD-OCT is illustrated in 17 eyes of 9 patients, including both the right (circle) and left (triangle) eyes. A total of 6 eyes had changes in disease stage during follow-up (dotted line). Overall, there is a trend toward increasing lesion diameter. **B,** Change in vitelliform lesion area on short-wavelength autofluorescence is illustrated in the same patients. Overall, there is a trend toward increasing lesion area. SW-AF = short-wavelength autofluorescence; SD-OCT = spectral-domain OCT.

quantify disease progression. To date, few studies describe the progression of BVMD on multimodal imaging, including those performed by Querques et al in 2014 and Parodi et al in 2019.[19,20] In this longitudinal study using multimodal imaging and cross-sectional data from DAC perimetry, we describe the changes that occur with disease progression in patients with BVMD.

The study by Querques et al provides longitudinal analysis of SD-OCT and SW-AF findings in 42 eyes of 21 patients.[19] The overall distribution of disease stages in their study was quite different from that seen in our study population, with 26% of eyes (11/42) in the previtelliform stage and 35% of eyes (15/42) having atrophic or fibrotic disease. In contrast, only 3% of eyes (1/30) and 7% of eyes (2/30) were found to have previtelliform and atrophic or fibrotic disease, respectively, in our cohort. In addition, our study featured 47% of eyes (14/30) with vitelliruptive lesions compared with 19% of eyes (8/42) in theirs. These

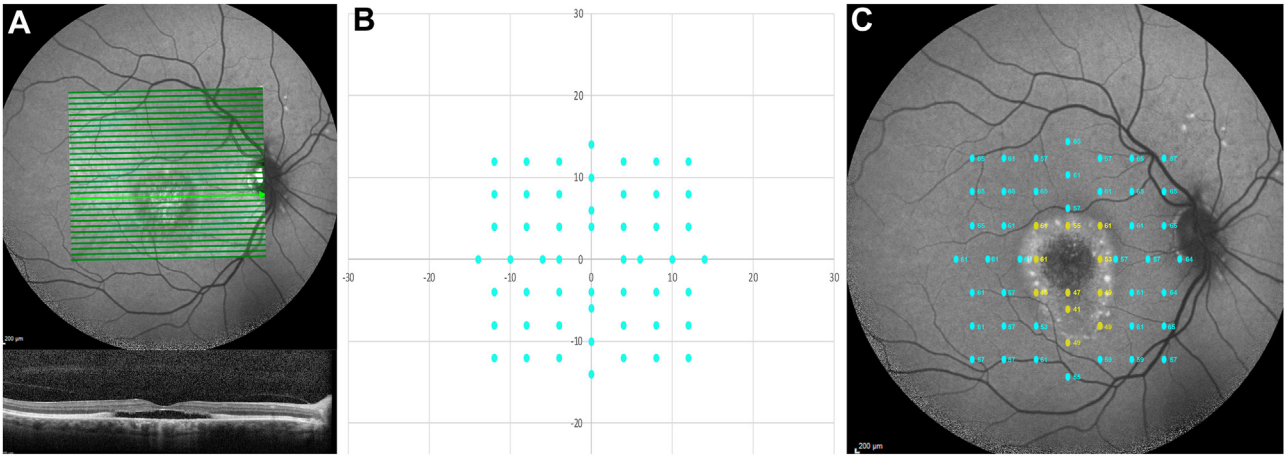


Figure 3. Rod-mediated sensitivity is affected in patients with Best vitelliform macular dystrophy. Patient 12 (P12). **A,** Near-infrared reflectance and spectral-domain OCT images were used to identify the fovea on a 55° SW-AF image. **B,** The 52 locations used to test scotopic sensitivity on the DAC perimeter are illustrated. **C,** The DAC perimeter cyan field numeric map in dBs inside (yellow) and outside (blue) the lesion was overlaid on the 55° SW-AF image. Scotopic sensitivities were decreased within the lesion, especially within the inferior aspect of the lesion. SW-AF = short-wavelength autofluorescence; DAC = dark-adapted chromatic; dBs = decibels.

differences may reflect patient demographics and age, as their study featured a bimodal distribution of disease stage, whereas our cohort demonstrates a unimodal left skew. Alternatively, the distribution of patients seen in our study may reflect our center's identity as a tertiary referral center, with many patients being referred for evaluation after already being diagnosed with significant disease.

Herein lies a unique opportunity to compare and contrast multimodal imaging in the nuanced context of disease staging. Similar to Querques et al and Parodi et al, we found that lesion thickness was the greatest in the pseudohypopyon stage, followed by a relative decrease in lesion thickness in the vitelliruptive stage and not surprisingly a greater decrease in the atrophic or fibrotic stage.[19,20] In our study, we identified a gradual increase in lesion diameter from the vitelliform through vitelliruptive stage, whereas Querques et al describe a similar trend to lesion thickness with peak lesion width identified in the pseudohypopyon rather than vitelliruptive stage.[19] This discrepancy may be confounded due to several patients seen in their cohort who reverted from the vitelliruptive stage to the pseudohypopyon stage and vice versa, a finding not observed in our cohort. More rapid growth of lesion width was seen across our cohort (105.4 µm/year vs. 12.4 µm/year) compared with that seen in the study by Querques et al.[19] This difference may reflect the greater incidence of disease progression seen in our cohort compared with that of Querques et al with 9 of 30 eyes demonstrating an overall progression in disease stage in our cohort compared with 3 of 42 eyes in their study. Duration of follow-up was also greater in our study (average 76.56 months) compared with theirs (average 38.88 months).[19]

Querques et al also described an average increase in CMT in their cohort (3.0 µm/year) compared with the average decrease in CMT seen in our cohort (−4.3 µm/year).[19] This may be due to the higher incidence of patients with vitelliruptive disease seen in our cohort, as these eyes approach atrophic disease, and the retina gradually thins. Parodi et al have similarly suggested that vitelliruptive disease is significantly associated with decreases in lesion, and by extension, retinal height.[20] Both our study and those of Querques et al and Parodi et al reported a progressive increase in the size of the lesion on SW-AF, but again, more rapid increases were seen in our population as compared with that of Querques et al (0.93 mm^2/year vs. 0.21 mm^2/year).[19,20]

A potential biomarker of disease progression in BVMD seen in our study is the measurement of ONL thickness along the temporal and nasal borders of the lesion as well as at 500 µm away from both the nasal and temporal borders (Fig 1). Analysis of retinal thickness measurement as a function of disease stage revealed significant differences. Prior studies by Cideciyan et al demonstrated a relative thickening of the ONL at some locations across the fovea in a total of 6 patients.[21] This is in contrast to our results, which show relative decreases in average thickness with progression of disease stage, which has also been described in previous studies.[26] Importantly, we previously demonstrated that the ONL thinning noted on SD-OCT additionally correlates with a halo surrounding the lesion seen on near-infrared autofluorescence and SW-AF, suggesting that changes in this feature can be identified in multimodal images.[25,27] Notably, a significant limitation of this measurement is the slow change in ONL thickness at the center of the lesion, temporal and nasal borders of the lesion, and 500 µm from the temporal and nasal borders of the lesion, with annual changes averaging <5 µm/year, consequently requiring long-term follow-up to identify utility. Future studies would benefit from tying these structural changes to fundus-guided perimetry for direct functional evaluation.

In addition to longitudinal imaging, cross-sectional analysis of visual function was tested using DAC

perimetry. Our results demonstrate that rod-mediated sensitivity is affected within the lesion area of patients with BVMD. Even though many patients had preserved central vision, DAC perimetry showed deficits in rod-mediated visual function. Specifically, rod-mediated sensitivity within the vitelliform lesion was lower compared with locations outside the lesion in the same eye, as well as compared with results for healthy eyes. Only 1 eye of the tested subjects had clear atrophic lesions that could account for the decrease in rod-mediated function; the majority of eyes that were tested were in the vitelliruptive or pseudohypopyon stage. This decrease in rod sensitivity may be explained by the subretinal fluid that is frequently seen in these stages of BVMD. Compared with cones, rod photoreceptors rely solely on RPE cells to regenerate visual pigment.[28] As such, the presence of subretinal fluid causing localized detachment of the photoreceptors from the RPE cells may retard the transfer of 11-*cis*-retinal from the RPE cells to the rods, thereby compromising the visual cycle. Our findings are in agreement with a recent study performed by Cideciyan et al demonstrating deterioration of rod rather than cone sensitivity using a modified Humphrey visual field analyzer.[21] Additionally, relatively lower scotopic sensitivities were noted in the inferior aspects of the lesions (Fig 2), which may be due to gravity-dependent pooling of subretinal fluid in the inferior lesion and therefore more interruption of 11-*cis*-retinal transfer. Longitudinal studies of DAC perimetry are needed; however, the cross-sectional data suggests that it may be a potentially sensitive outcome measure of disease.

While this study provides a unique comparison to different cohorts from previously published studies, it is limited by several factors, including sample size and follow-up.[19,20] For use in clinical trials, outcome measures such as lesion width and retinal thickness may increase or decrease depending on clinical stage and often in a nonlinear fashion. Despite a follow-up period of >5 years, several measurements of change occurred at increments of <5 μm per year and are consequently subject to the variability associated with the resolution of the OCT scan (roughly 3–7 μm) and reader measurement.[29,30] Given the significant difficulty associated with establishing large cohorts of BVMD patients to document the natural history of BVMD disease, multicenter collaboration will be necessary to elucidate outcome measures.

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No animal subjects were used in this study.

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Abbreviations and Acronyms:
BVMD = Best vitelliform macular dystrophy; **CMT** = central macular thickness; **DAC** = dark-adapted chromatic; **dB** = decibel; **ONL** = outer

nuclear layer; **RPE** = retinal pigment epithelium; **SW-AF** = short-wavelength autofluorescence; **SD-OCT** = spectral-domain OCT.

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